

Protein intake, calcium balance and health consequences.

High-protein (HP) diets exert a hypercalciuric effect at constant levels of calcium intake, even though the effect may depend on the nature of the dietary protein. Lower urinary pH is also consistently observed for subjects consuming HP diets. The combination of these two effects was suspected to be associated with a dietary environment favorable for demineralization of the skeleton. However, increased calcium excretion due to HP diet does not seem to be linked to impaired calcium balance. In contrast, some data indicate that HP intakes induce an increase of intestinal calcium absorption. Moreover, no clinical data support the hypothesis of a detrimental effect of HP diet on bone health, except in a context of inadequate calcium supply. In addition, HP intake promotes bone growth and retards bone loss and low-protein diet is associated with higher risk of hip fractures. The increase of acid and calcium excretion due to HP diet is also accused of constituting a favorable environment for kidney stones and renal diseases. However, in healthy subjects, no damaging effect of HP diets on kidney has been found in either observational or interventional studies and it seems that HP diets might be deleterious only in patients with preexisting metabolic renal dysfunction. Thus, HP diet does not seem to lead to calcium bone loss, and the role of protein seems to be complex and probably dependent on other dietary factors and the presence of other nutrients in the diet.